INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

# THRUSTER, CARTRIDGE ACTUATED, M25A1 METAL PARTS ASSEMBLY

*This specification is approved for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the M25A1 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

## 2. APPLICABLE DOCUMENTS

### 2.1 Government documents.

**2.1.1 Specifications and standards.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-A-2550 | Ammunition, General Specification For |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-413 | Visual Inspection Guide for Elastomeric O-rings |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)



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**2.1.2 Other Government documents, drawings, and publications.** The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

| | |
|---|---|
| 11730535 | Thruster, Cartridge Actuated, M25A1 Metal Parts Assembly |
| 8797741 | Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1 |
| 8797742 | Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1 |
| 9217579 | Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1 |

U.S. Air Force

| | |
|---|---|
| TPO 00-554-3103 | Thruster, Cartridge Actuated, M25A1 |

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

**2.2 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

**3.2 Materials.** All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

**3.2.1 Raw material.** The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.



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**3.3 Hardness and other physical properties.** Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

**3.4 Hydrostatic pressure.** The body and head shall withstand a hydrostatic pressure of 8,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

**3.5 Structural strength.** The assembled thruster shall withstand a static tensile force of 2,000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.4.1).

**3.6 Shear pin strength.** The shear pin, after application of protective finish, shall shear when a load of 48 ± 4 pounds is applied at the steady rate of 0.125 inch per minute.

**3.7 Metal parts assembly.** Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

**3.8 Protective finish.** Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

**3.9 Residual magnetism.** The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.6.

**3.10 Workmanship.** Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

**3.10.1 Metal defects.** All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

**3.10.2 Burr.** No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

**3.10.3 Foreign matter.** No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).



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**4.1.1 Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

**4.1.2 Quality assurance terms and definitions.** Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

**4.1.3 Inspections.** Inspections shall be in accordance with MIL-A-2550.

**4.2 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

**4.3 Inspection conditions.** Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

**4.4 First article inspection.** At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

**4.5 Quality conformance inspection.**

**4.5.1 Lot.**

**4.5.1.1 Submission of product.** Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

**4.5.1.2 Size of lot.** The production lot size shall be as specified in the contract.



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**4.5.2 Examination.** One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

**4.5.2.1 Key, Drawing 10534180, covering a detail of Drawing 11730535.**

| Characteristics | Defect | Method of Inspection |
|---|---|---|
| Critical | | |
| None defined | | |
| Major | | |
| 101 | Surface finish incorrect | Visual |
| 102 | Protective finish inadequate or defective | Visual |
| 103 | Metal defective | Visual |
| 104 | Burr or sharp corner | Visual |
| 105 | Foreign matter | Visual |
| Minor | | |
| 201 | Total length | Gage |
| 202 | Length from end to vertical center | Gage |
| 203 | Length from top to true position of longitudinal center | Gage |
| 204 | Length from bottom of true position of longitudinal center | Gage |
| 205 | Width | Gage |
| 206 | Height | Gage |
| 207 | Radius or chamfer incorrect | Gage |
| 208 | Radius or chamfer missing | Visual |

**4.5.2.2 Pin, Firing, Drawing 8593864, covering a detail of Drawing 11730535.**

| | | |
|---|---|---|
| Critical | | |
| None defined | | |
| Major | | |
| 101 | Concentricity of groove with body | Gage |
| 102 | Concentricity of striker with body | Gage |
| 103 | Concentricity of striker shoulder with body | Gage |
| 104 | True position of hole | Gage |
| 105 | Diameter of groove | Gage |
| 106 | Diameter of body, min | Gage |
| 107 | Diameter of striker, max | Gage |
| 108 | Diameter of hole, max | Gage |
| 109 | Profile of striker | Gage |
| 110 | Profile of groove | Gage |
| 111 | Metal defective | Visual |



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| | | |
|---|---|---|
| 112 | Protective finish inadequate or defective | Visual |
| 113 | Surface finish of body and striker tip incorrect | Visual |
| 114 | Burr or sharp corner | Visual |
| 115 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Diameter of body, max | Gage |
| 202 | Diameter of striker, min | Gage |
| 203 | Diameter of hole, min | Gage |
| 204 | Diameter of striker shoulder | Gage |
| 205 | Depth of hole | Gage |
| 206 | Length from body to tip of striker | Gage |
| 207 | Length of striker | Gage |
| 208 | Total length | Gage |
| 209 | Location of hole | Gage |
| 210 | Location of groove | Gage |
| 211 | Depth of slot in base | Gage |
| 212 | Length of body diameter forward of groove, min | Gage |
| 213 | Width of groove | Gage |
| 214 | Width of slot in base | Gage |
| 215 | Location of slot in base | Gage |
| 216 | Radius incorrect | Gage |
| 217 | Radius missing | Visual |
| 218 | Surface finish improper, except as otherwise classified | Visual |
| 219 | Marking missing, incorrect, or illegible | Visual |

#### 4.5.2.3 Body Assembly, Drawing 10534179, covering a detail of Drawing 11730535.

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Pitch diameter of thread, min | Gage |
| 102 | Major diameter of thread, min | Gage |
| 103 | Concentricity of pitch diameter of thread with cavity | Gage |
| 104 | Concentricity of bore with cavity | Gage |
| 105 | Concentricity of groove with bore | Gage |
| 106 | Concentricity of small counterbore with bore | Gage |
| 107 | Concentricity of large counterbore with bore | Gage |
| 108 | True position of slots | Gage |
| 109 | Symmetry of slots with each other and body datum diameter | Gage |
| 110 | Diameter of thread undercut, min | Gage |
| 111 | Diameter of cavity, max | Gage |
| 112 | Diameter of internal undercut, max (if applicable) | Gage |
| 113 | Diameter of bore | Gage |
| 114 | Diameter of groove | Gage |
| 115 | Diameter of small counterbore | Gage |
| 116 | Diameter of body at threaded end | Gage |
| 117 | Location of groove | Gage |



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| | | |
|---|---|---|
| 118 | Protrusion of pellet, max | Gage |
| 119 | Straightness of cavity | Gage |
| 120 | Width of groove | Gage |
| 121 | Profile of groove | Gage |
| 122 | Thread damaged or not full | Visual |
| 123 | Metal defective | Visual |
| 124 | Protective finish inadequate or defective | Visual |
| 125 | Surface finish of cavity improper | Visual |
| 126 | Burr or sharp corner | Visual |
| 127 | Foreign matter | Visual |
| 128 | Pellet missing, loose, or improperly seated | Visual |
| Minor | | |
| 201 | Pitch diameter of thread, max | Gage |
| 202 | Diameter of cavity, min | Gage |
| 203 | Diameter of large counterbore | Gage |
| 204 | Diameter of body, max | Gage |
| 205 | Width of flat at threaded face | Gage |
| 206 | Thickness of wall | Gage |
| 207 | Length from end to shoulder of large counterbore | Gage |
| 208 | Length from end to large shoulder of small counterbore | Gage |
| 209 | Total length | Gage |
| 210 | Width across flats | Gage |
| 211 | Depth of slots | Gage |
| 212 | Width of slots | Gage |
| 213 | Location of pellets from tapered face | Gage |
| 214 | Length from front tapered face to exterior shoulder | Gage |
| 215 | Diameter of pellet hole | Gage |
| 216 | Depth of pellet hole | Gage |
| 217 | Bottom of pellet hole not flat | Gage |
| 218 | Width of thread undercut, max | Gage |
| 219 | Width of interior undercut, max | Gage |
| 220 | Length from end to inside front face of bore | Gage |
| 221 | Diameter of exterior shoulder | Gage |
| 222 | Width of exterior shoulder | Gage |
| 223 | Location of undercut from shoulder | Gage |
| 224 | Location of flats incorrect | Gage |
| 225 | Pellet protrusion beyond major thread diameter, max | Gage |
| 226 | Radius or chamfer incorrect | Gage |
| 227 | Radius or chamfer missing | Visual |
| 228 | Surface finish improper, except as otherwise specified | Visual |
| 229 | Marking missing, incorrect, or illegible | Visual |

#### 4.5.2.4 Trunnion, Drawing 8594482, covering a detail of Drawing 11730535.

Critical
None defined



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Major

| | | |
|---|---|---|
| 101 | Concentricity of lugs | Gage |
| 102 | True position of lugs | Gage |
| 103 | Symmetry of flats at root of lugs with inside diameter | Gage |
| 104 | Perpendicularity of sides with inside diameter | Gage |
| 105 | Diameter of lugs, min | Gage |
| 106 | Width across flats, min | Gage |
| 107 | Radius at base of lugs | Gage |
| 108 | Metal defective | Visual |
| 109 | Protective finish inadequate or defective | Visual |

Minor

| | | |
|---|---|---|
| 201 | Diameter of lugs, max | Gage |
| 202 | Outside diameter, max | Gage |
| 203 | Inside diameter | Gage |
| 204 | Length of lugs | Gage |
| 205 | Width across flats, max | Gage |
| 206 | Width of body | Gage |
| 207 | Thickness of wall | Gage |
| 208 | Location of lugs | Gage |
| 209 | Radius or chamfer incorrect, except as otherwise classified | Gage |
| 210 | Radius or chamfer missing | Visual |
| 211 | Surface finish improper | Visual |
| 212 | Burr or sharp corner | Visual |
| 213 | Foreign matter | Visual |
| 214 | Marking missing, incorrect, or illegible | Visual |

#### 4.5.2.5 Spring, Piston Locking, Drawing 10534184, covering a detail of Drawing 11730535.

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Load at specified height | Gage |
| 102 | Solid length, max | Gage |
| 103 | Ends incorrect | Visual |
| 104 | Protective finish inadequate or defective | Visual |
| 105 | Burr | Visual |

Minor
None defined

#### 4.5.2.6 Head, Drawing 10521455, covering a detail of Drawing 11730535.

Critical
None defined

Major



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| | | |
|---|---|---|
| 101 | Pitch diameter of small thread, max | Gage |
| 102 | Minor diameter of small thread, max | Gage |
| 103 | Pitch diameter of midsection thread, max | Gage |
| 104 | Minor diameter of midsection thread, max | Gage |
| 105 | Pitch diameter of large thread, max | Gage |
| 106 | Minor diameter of large thread, max | Gage |
| 107 | Concentricity of pitch diameter of large thread with pitch diameter of midsection thread | Gage |
| 108 | Depth of cavity, small end | Gage |
| 109 | Depth of counterbore, small end | Gage |
| 110 | Length of thread, small end, max | Gage |
| 111 | Length of effective thread, small end, min | Gage |
| 112 | Diameter of undercut, large thread, max | Gage |
| 113 | Diameter of undercut, midsection thread, max | Gage |
| 114 | Diameter of chamfer at small end | Gage |
| 115 | Angle of chamfer at small end | Gage |
| 116 | Thread damaged or not full | Visual |
| 117 | Metal defective | Visual |
| 118 | Protective finish inadequate or defective | Visual |
| 119 | Surface finish of face in midsection cavity, chamfers in small end and small thread face improper | Visual |
| 120 | Burr | Visual |
| 121 | Foreign matter | Visual |
| | | |
| Minor | | |
| 201 | Pitch diameter of small thread, min | Gage |
| 202 | Pitch diameter of midsection thread, min | Gage |
| 203 | Pitch diameter of large thread, min | Gage |
| 204 | Diameter of counterbore, small end | Gage |
| 205 | Diameter of hole through midsection | Gage |
| 206 | Depth of large cavity | Gage |
| 207 | Diameter of body, midsection | Gage |
| 208 | Diameter of body, large end | Gage |
| 209 | Diameter of body, small end | Gage |
| 210 | Depth to bottom of midsection | Gage |
| 211 | Length of large body diameter | Gage |
| 212 | Length of small body diameter | Gage |
| 213 | Length of flat | Gage |
| 214 | Total length | Gage |
| 215 | Width of undercut, large thread | Gage |
| 216 | Width of undercut, midsection thread | Gage |
| 217 | Width across flats | Gage |
| 218 | Location of flats | Gage |
| 219 | Radius or chamfer incorrect, except as otherwise classified | Gage |
| 220 | Radius or chamfer missing | Visual |
| 221 | Surface finish improper, except as otherwise classified | Visual |
| 222 | Marking missing, incorrect, or illegible | Visual |

**4.5.2.7 Piston, Drawing 10534181, covering a detail of Drawing 11730535.**



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Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Pitch diameter of thread, min | Gage |
| 102 | Major diameter of thread, min | Gage |
| 103 | Pitch diameter of small threads, max | Gage |
| 104 | Minor diameter of small threads, max | Gage |
| 105 | Concentricity of small thread holes minor diameter | Gage |
| 106 | Concentricity of body midsection with large end | Gage |
| 107 | Concentricity of shoulder with large end | Gage |
| 108 | Concentricity of groove with body, large end | Gage |
| 109 | Concentricity of large cavity with large outside shoulder | Gage |
| 110 | Concentricity of small cavity with large cavity | Gage |
| 111 | Concentricity of intermediate cavity with large cavity | Gage |
| 112 | Symmetry of slots | Gage |
| 113 | True position of holes with body | Gage |
| 114 | Diameter of groove | Gage |
| 115 | Diameter of body at large end, min | Gage |
| 116 | Diameter of body between wrench flats and undercut, min | Gage |
| 117 | Diameter of large cavity | Gage |
| 118 | Location of slots | Gage |
| 119 | Diameter of intermediate cavity | Gage |
| 120 | Location of large shoulder of intermediate cavity | Gage |
| 121 | Profile of groove | Gage |
| 122 | Thread damaged or not full | Visual |
| 123 | Metal defective | Visual |
| 124 | Protective finish inadequate or defective | Visual |
| 125 | Surface finish improper | Visual |
| 126 | Burr or sharp corner | Visual |
| 127 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Pitch diameter of thread, max | Gage |
| 202 | Diameter of body at large end, max | Gage |
| 203 | Diameter of shoulder, large end | Gage |
| 204 | Pitch diameter of small thread, min | Gage |
| 205 | Diameter of drill point | Gage |
| 206 | Diameter of undercut, min (if applicable) | Gage |
| 207 | Diameter of small cavity | Gage |
| 208 | Length of body at large end | Gage |
| 209 | Depth of holes | Gage |
| 210 | Length of slots | Gage |
| 211 | Width of slots | Gage |
| 212 | Profile of slots | Gage |
| 213 | Length of wrench flats | Gage |
| 214 | Length of shoulder, large end | Gage |
| 215 | Location of holes | Gage |
| 216 | Diameter of large cavity, min | Gage |



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| | | |
|---|---|---|
| 217 | Length from shoulder to small end of body | Gage |
| 218 | Depth of large cavity | Gage |
| 219 | Total length | Gage |
| 220 | Diameter of face | Gage |
| 221 | Width of undercut, max (if applicable) | Gage |
| 222 | Location of groove | Gage |
| 223 | Width across wrench flats | Gage |
| 224 | Length of thread | Gage |
| 225 | Radius or chamfer incorrect | Gage |
| 226 | Radius or chamfer missing | Visual |
| 227 | Surface finish improper, except as otherwise classified | Visual |
| 228 | Marking missing, incorrect, or illegible | Visual |
| 229 | Diameter (2) holes through wrench flats | Gage |
| 230 | Location (2) holes through wrench flats | Gage |

**4.5.2.8 Guide Retainer Assembly, Drawing 10521616, covering a detail of Drawing 11730535.**

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Pitch diameter of external thread, min | Gage |
| 102 | Major diameter of external thread, min | Gage |
| 103 | Pitch diameter of internal thread, max | Gage |
| 104 | Minor diameter of internal thread, max | Gage |
| 105 | Concentricity of large cavity with pitch diameter of internal thread | Gage |
| 106 | Concentricity of boss with pitch diameter of internal thread | Gage |
| 107 | Concentricity of body with pitch diameter of external thread | Gage |
| 108 | Concentricity of small cavity with pitch diameter of internal thread | Gage |
| 109 | Concentricity of groove with body | Gage |
| 110 | Concentricity of hole through boss with pitch diameter of internal thread and small cavity | Gage |
| 111 | Concentricity of recess with small cavity | Gage |
| 112 | Perpendicularity of boss face and base of small cavity with pitch diameter of internal thread | Gage |
| 113 | True position of hole | Gage |
| 114 | Diameter of groove | Gage |
| 115 | Diameter of small cavity | Gage |
| 116 | Diameter of large cavity, max | Gage |
| 117 | Diameter of hole through boss | Gage |
| 118 | Diameter of body | Gage |
| 119 | Diameter of undercut, max | Gage |
| 120 | Height of boss | Gage |
| 121 | Thickness through boss | Gage |
| 122 | Thickness of base of small cavity | Gage |
| 123 | Protrusion of pellets, max | Gage |
| 124 | Profile of groove | Gage |



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| | | |
|---|---|---|
| 125 | Surface finish in small cavity improper | Visual |
| 126 | Metal defective | Visual |
| 127 | Protective finish inadequate or defective | Visual |
| 128 | Thread damaged or not full | Visual |
| 129 | Pellets loose, missing, or improperly inserted | Visual |
| 130 | Burr or sharp corner | Visual |
| 131 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Pitch diameter of external thread, max | Gage |
| 202 | Pitch diameter of internal thread, min | Gage |
| 203 | Symmetry of flats with pitch diameter of internal thread | Gage |
| 204 | Diameter of recess at base of small cavity | Gage |
| 205 | Diameter of large cavity, min | Gage |
| 206 | Diameter at rear of external thread | Gage |
| 207 | Diameter of pin hole | Gage |
| 208 | Diameter of pellet hole | Gage |
| 209 | Depth of large cavity | Gage |
| 210 | Location of flange | Gage |
| 211 | Total length | Gage |
| 212 | Width across flats | Gage |
| 213 | Location of hole | Gage |
| 214 | Location of groove | Gage |
| 215 | Flatness at bottom of pellet hole | Gage |
| 216 | Depth of pellet hole | Gage |
| 217 | Location of undercut | Gage |
| 218 | Location of pellets | Gage |
| 219 | Width of groove | Gage |
| 220 | Length of minimum effective groove | Gage |
| 221 | Width of undercut at internal thread | Gage |
| 222 | Length of flats | Gage |
| 223 | Radius or chamfer incorrect | Gage |
| 224 | Surface finish improper, except as otherwise classified | Visual |
| 225 | Marking missing, incorrect, or illegible | Visual |

#### 4.5.2.9 Sleeve Assembly, Drawing 10521617, covering a detail of Drawing 11730535.

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Pitch diameter of thread, min | Gage |
| 102 | Major diameter of thread, min | Gage |
| 103 | Concentricity of bore with pitch diameter of thread | Gage |
| 104 | Diameter of bore | Gage |
| 105 | Depth of pellet hole prior to pellet insertion | Gage |



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| | | |
|---|---|---|
| 106 | Diameter of pellet hole prior to pellet insertion | Gage |
| 107 | Location of pellet hole | Gage |
| 108 | Pellet below flush with major diameter | Gage |
| 109 | Protrusion of pellet, max | Gage |
| 110 | Radius at end of bore improper | Visual |
| 111 | Thread damaged or not full | Visual |
| 112 | Metal defective | Visual |
| 113 | Protective finish inadequate or defective | Visual |
| 114 | Surface finish of radius at end of bore improper | Visual |
| 115 | Burr | Visual |
| 116 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Pitch diameter of thread, max | Gage |
| 202 | Symmetry of flats with pitch diameter of thread | Gage |
| 203 | Width across flats | Gage |
| 204 | Total length | Gage |
| 205 | Length of thread | Gage |
| 206 | Length of flats | Gage |
| 207 | Radius or chamfer incorrect | Gage |
| 208 | Radius or chamfer missing, except as otherwise classified | Visual |
| 209 | Surface finish improper, except as otherwise classified | Visual |
| 210 | Marking missing, incorrect, or illegible | Visual |

#### 4.5.2.10 Pin, Shear, Drawing 10521618, covering a detail of Drawing 11730535.

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Diameter, max | Gage |
| 102 | Metal defective | Visual |
| 103 | Protective finish inadequate or defective | Visual |
| 104 | Burr | Visual |
| 105 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Total length | Gage |
| 202 | Profile of end | Visual |

#### 4.5.2.11 Piston, Unlocking, Drawing 10534182, covering a detail of Drawing 11730535.

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Concentricity of groove with large outside diameter | Gage |
| 102 | Concentricity of body with large outside diameter | Gage |



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| | | |
|---|---|---|
| 103 | Concentricity of cavity with large outside diameter | Gage |
| 104 | Concentricity of small diameter at cavity end with large outside diameter | Gage |
| 105 | Concentricity of immediate diameter at cavity end with large outside diameter | Gage |
| 106 | Concentricity of large diameter at cavity end with large outside diameter | Gage |
| 107 | Diameter at large end, min | Gage |
| 108 | Diameter of groove, min | Gage |
| 109 | Length from face at cavity end to taper, max | Gage |
| 110 | Depth of cavity | Gage |
| 111 | Width of groove, max | Gage |
| 112 | Profile of groove | Gage |
| 113 | Chamfer improper or incorrect | Visual |
| 114 | Protective finish inadequate or defective | Visual |
| 115 | Burr or sharp corner | Visual |
| 116 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Diameter of large end, max | Gage |
| 202 | Diameter of groove, max | Gage |
| 203 | Diameter of body | Gage |
| 204 | Diameter of cavity | Gage |
| 205 | Small diameter at cavity end | Gage |
| 206 | Immediate diameter at cavity end | Gage |
| 207 | Large diameter at cavity end | Gage |
| 208 | Diameter of drill point, max | Gage |
| 209 | Angle at tapered surface | Gage |
| 210 | Width of groove, min | Gage |
| 211 | Thickness of lip | Gage |
| 212 | Length from face at cavity end to taper, min | Gage |
| 213 | Total length | Gage |
| 214 | Length from front face to body | Gage |
| 215 | Location of groove | Gage |
| 216 | Length from face of cavity end to body | Gage |
| 217 | Radius or chamfer missing, except as specified | Visual |
| 218 | Marking missing, incorrect, or illegible | Visual |
| 219 | Surface finish improper | Visual |

#### 4.5.2.12 Ring, Drawing 10534183, covering a detail of Drawing 11730535.

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Concentricity of exterior diameter with internal diameter | Gage |
| 102 | Concentricity of small holes | Gage |
| 103 | Concentricity of center bore with small holes | Gage |



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| | | |
|---|---|---|
| 104 | Chamfers at exterior and interior surfaces | Visual |
| 105 | Protective finish inadequate or defective | Visual |
| 106 | Burr or sharp corner | Visual |
| 107 | Foreign matter | Visual |

Minor

| | | |
|---|---|---|
| 201 | Diameter at exterior surface | Gage |
| 202 | Diameter at interior surface | Gage |
| 203 | Diameter of holes | Gage |
| 204 | Diameter of counterbore | Gage |
| 205 | Depth of counterbore | Gage |
| 206 | Length from front face to rear face | Gage |
| 207 | Location of holes | Gage |
| 208 | Radius or chamfer incorrect | Gage |
| 209 | Radius or chamfer missing | Visual |
| 210 | Surface finish improper | Visual |
| 211 | Marking missing, incorrect, or illegible | Visual |

**4.5.2.13 Assembly, Drawing 11730535, less all components except 10521616, 8593864, and MS29513-12.**

Critical

| | | |
|---|---|---|
| 1 | Firing pin protrusion | Gage |

Major

| | | |
|---|---|---|
| 101 | Any part missing | Visual |
| 102 | Protective finish inadequate or defective | Visual |
| 103 | Burr | Visual |
| 104 | Foreign matter | Visual |

Minor
None defined

**4.5.2.14 Assembly, Drawing 11730535.**

Critical
None defined

Major

| | | |
|---|---|---|
| 101 | Trunnion not free to rotate 360 degrees | Manual |
| 102 | Any part missing, except as otherwise classified | Visual |
| 103 | Improper assembly | Visual |
| 104 | Thread damaged or not full | Visual |
| 105 | Marking missing,incorrect, or illegible | Visual |
| 106 | Burr | Visual |
| 107 | Foreign matter | Visual |



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Minor

| | | |
|---|---|---|
| 201 | Protective cap or plug missing | Visual |
| 202 | Protective finish inadequate or defective | Visual |

**4.5.2.15 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8797741.**

Critical
None defined

Major
None defined

Minor

| | | |
|---|---|---|
| 201 | Contents loose | Manual |
| 202 | Box damaged | Visual |
| 203 | Marking missing, incorrect, or illegible | Visual |
| 204 | Cut, damaged, or improperly sealed bag | Visual |

**4.5.2.16 Box, Packing, Ammunition, Drawing 8797742.**

Critical
None defined

Major
None defined

Minor

| | | |
|---|---|---|
| 201 | Contents loose | Visual/Manual |
| 202 | Strapping missing, loose, broken, or improperly assembled | Visual/Manual |
| 203 | Contents of box exposed | Visual |
| 204 | Box damaged | Visual |
| 205 | Marking missing, incorrect, or illegible | Visual |
| 206 | Number of units incorrect | Visual |

**4.5.2.17 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217579.**

Critical
None defined

Major
None defined

Minor

| | | |
|---|---|---|
| 201 | Contents loose | Manual/Visual |
| 202 | Carton broken or damaged | Visual |
| 203 | Number of units incorrect | Visual |
| 204 | Marking missing, incorrect, or illegible | Visual |
| 205 | Tape incomplete | Visual |



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**4.5.3 Inspection testing.** The following tests shall be performed in accordance with 4.6.

**4.5.3.1 Hardness.** Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

**4.5.3.2 Hydrostatic pressure.** Each body and head shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.4 shall be cause for rejection of the subassembly.

**4.5.3.3 Structural strength.** Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.5 shall be cause for rejection of the thruster.

**4.5.3.4 Shear pin strength.** Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.6 shall be cause for rejection of the lot.

**4.5.3.5 Protective finish.** The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

**4.5.3.6 Residual magnetism.** Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.6. Failure of any unit in the inspection sample to comply with 3.9 shall be cause for rejection of the assembly.

**4.6 Test methods and procedures.**

**4.6.1 Hardness.** Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

**4.6.2 Hydrostatic pressure.** Prior to application of protective finish the body and end cap shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

**4.6.3 Structural strength.** The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.5.

**4.6.4 Shear pin strength.** After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

**4.6.5 Protective finish.** Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

**4.6.6 Residual magnetism.** The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and



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moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

**4.7 Inspection of packaging.** The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

## 5. PACKAGING

**5.1 Packing and marking.** Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8797741 and 8797742. Level B shall be in accordance with Drawings 8797741 and 9217579.

**5.2 Air Force packaging.** Air Force packaging shall be in accordance with TPO 00-554-3103.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

**6.1 Intended use.** The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

**6.3 First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.



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**6.4 Definition.**

**6.4.1 Mechanical failure.** Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

**6.5 Subject term (keyword) listing.**

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

**6.6 Changes from previous issue.** Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E02)

Review activity:
Air Force - 70



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